PROJECT BRIEF / FULL-STACK ENGINEERING

DormFix

A shared workflow for landlords and tenants.

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PROBLEM & USERS

Manual logbooks and scattered messages make it difficult to track tenant approval, room capacity, submitted payments, and unresolved repairs. Landlords and tenants need a consistent view of those operations.

Built for: Landlords · tenants

MY CONTRIBUTION

My classmate proposed the capstone; I implemented its features as part of the team.

MAIN TECHNOLOGIES

React 19 / TypeScript / Node.js / Express 5 / MSSQL / Tesseract.js / Ollama LLaVA / Zod / Socket.io / Tailwind CSS / DaisyUI

VERSION SCOPE

Historical OCR and local-model design. This case study follows the earlier Tesseract, Ollama, Zod-defined response format, and SQL Server design. Later revisions use Gemini and PostgreSQL; the evolution timeline traces those changes.

WORKFLOW

01. Pair and approve

02. Assign a room

03. Review a payment

04. Resolve issues

SYSTEM STRUCTURE

Interface: React / role-based views

API: Express / TypeScript

Receipt analysis: Tesseract > Ollama > Zod

Data & events: MSSQL / Socket.io

ENGINEERING DECISION

Separate OCR and local model parsing from business checks. Use a Zod-defined response format, then compare extracted fields with submitted payment details; the reviewed parser does not independently validate its parsed JSON.

TRADE-OFF

The historical local pipeline avoids a hosted model API fee but needs server compute. A requested response shape does not guarantee accurate extraction; business checks and landlord review remain necessary.

READ CASE STUDY | TRY WALKTHROUGH | VIEW SOURCE

Portfolio case studies and walkthroughs provide further detail and sample interactions.